



Accumulation of Opportunities Predicts the Educational Attainment and Adulthood Earnings of Children Born Into Low- Versus Higher-Income Households

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Scholars theorize that “opportunity gaps” drive achievement disparities between children born into poverty versus affluence. In a 26-year longitudinal study ($N = 814$), we examine (a) economic disparity in children’s accumulation of opportunities—from birth through high school—at home, childcare, school, afterschool, and in the neighborhood; (b) the extent to which opportunity is linked with educational attainment and earnings in early adulthood; and (c) whether opportunity is most strongly associated with these adult outcomes for children from low-income households. We document large opportunity gaps between children from low- versus higher-income households. These opportunity gaps are strongly linked with educational attainment and earnings, particularly for low-income children, helping explain why household income in early childhood predicts these outcomes in adulthood.

Keywords: achievement gap; child development; disparities; equity; human development; longitudinal studies; observational research; poverty; regression analyses

Poverty in early childhood is robustly linked with low educational attainment and low earnings in adulthood (e.g., National Academies of Sciences, Engineering, and Medicine [NASEM], 2023b). In efforts to understand how these lasting consequences unfold in children’s lives, education researchers have increasingly rejected explanations focused on deficits within poor families and narratives that ignore the larger systemic and environmental context in which children develop. Instead, scholars have theorized that “opportunity gaps”—systemic, pervasive, and compounding disparities in access to resources that support learning and development across childhood and adolescence—are to blame (Carter & Welner, 2013; Gorski, 2017; NASEM, 2023a).

Evidence for the opportunity gap perspective is sizable but largely based on empirical snapshots of children’s lives in one type of setting (e.g., at home *or* in the classroom *or* in their neighborhoods) at a single stage of life (e.g., during early childhood *or* middle childhood *or* adolescence). The present study is the first to directly measure the longitudinal accumulation of opportunities across settings and across children’s lives, from birth through the end of high school, using assessments of five

key settings: the home, early education and care, elementary school classrooms, organized afterschool activities, and neighborhoods. Following a sample of children born into lower-, middle-, and higher-income households for 26 years, we document opportunities and test the hypothesis that cumulative opportunities help explain the lasting consequences of household income in early childhood for educational attainment and earnings in early adulthood.

Theoretical Perspectives on Cumulative Opportunities

Key to cumulative opportunity gap perspectives in education research is the argument that “achievement gaps” are a predictable outcome of compounding disadvantage and harm that have accrued for children in poverty due to generations of sociopolitical and economic underinvestment in their livelihoods

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(Darling-Hammond, 2013; Ladson-Billings, 2006). Indeed, there is overwhelming evidence that within contexts of poverty, a constellation of societal forces works to disadvantage children, families, schools, and communities via segregation, disinvestment, and policy favoring the economically advantaged (e.g., Durlauf et al., 2022; Johnson, 2014; Owens, 2018; Reardon & Bischoff, 2011). Scholars argue that the result is severe disparity in children's opportunities to accumulate high-quality learning experiences, both in and out of school, as a function of family economic standing (Darling-Hammond, 2013). Importantly, cumulative opportunity gap perspectives are a rejection of the "culture of poverty" thesis that permeated education and sociological research for several decades (Demos, 1990; Gorski, 2008). On the other hand, the proposal that cumulative opportunity gaps are the driver of achievement gaps is highly consistent with leading bioecological systems and developmental systems theories of child development (Bronfenbrenner & Morris, 2007; Lerner & Castellino, 2002).

Developmental systems theorists argue that individual differences in learning and achievement are driven, in large part, by proximal processes in settings that directly contain the child, including learning experiences in their homes, childcare and early education centers, schools, afterschool settings, and neighborhoods (Bronfenbrenner & Morris, 2007; Lerner & Castellino, 2002). The impact of these experiences is, in turn, partly determined by their accumulation across settings, for example, meso-system processes operating across microsystems (e.g., connections between home and school environments) in bioecological systems theory or the integration of relational experiences across levels of organization in developmental systems theory (Bronfenbrenner & Morris, 2007; Lerner & Castellino, 2002).

Similarly, cognitive development theory emphasizes the importance of exposure to a diverse array of enriching environments for learning and brain growth (Greenough & Black, 2013; Piaget & Cook, 1952), and life course theory emphasizes the importance of repeated exposure to opportunity for building hope, self-efficacy, and volition (Elder et al., 2003). Moreover, the odds of children finding their niche—person-environment goodness of fit—likely increase in proportion to the number of supportive developmental contexts they experience. This is a critical point given that developmental theorists have long underscored the importance of niche picking, a process by which children select into and are then reinforced by environmental supports that are aligned with their developmental strengths and talents (Scarr & McCartney, 1983). Empirically, however, we lack a comprehensive portrait of the unfolding of opportunities as children grow, navigating social milieu and educational environs.

A Robust but Piecemeal Empirical Literature on Cumulative Opportunities

Although there is considerable evidence consistent with an opportunity gap perspective, this evidence is largely piecemeal (NASEM, 2023a). The vast majority of studies are limited to either one setting (e.g., home or preschool or school or afterschool or neighborhood) or one stage (e.g., early childhood or middle childhood or adolescence) in a child's life. Consider, for

example, studies of family life at the low and high ends of the income distribution. Having grown out of an array of theories on human capital and family investment, bioecological systems, social capital, cultural capital, and others (Becker & Thomes, 1986; Bronfenbrenner & Morris, 2007; Dubos, 2017; Yosso, 2005), the resulting empirical literature is robust and comprehensive, and it is heavily cited in opportunity gap arguments (e.g., NASEM, 2023a). With notable exceptions (e.g., Crosnoe et al., 2010; Owens, 2010), however, research has been fragmented with studies of opportunities in the home largely distinct from studies of opportunities outside the home.

Studies, for example, have focused on opportunities in early childhood education and care (e.g., Duncan & Magnuson, 2013) or schooling (e.g., Owens & Candipan, 2019) or afterschool (e.g., Vandell et al., 2015) or neighborhoods (e.g., Chetty & Hendren, 2018). Indeed, we have extensive evidence of disparity between children from low- versus high-income households in (a) quality of the home learning environment (e.g., Bradley et al., 2001) or (b) access to high-quality childcare and early education (e.g., Banghart et al., 2020) or (c) effective classroom instructional practices in elementary school (e.g., Peske & Haycock, 2006) or (d) access to structured afterschool activities (e.g., Simpkins et al., 2005) or (e) advantaged neighborhood conditions (e.g., Leventhal & Dupéré, 2019). Moreover, we know that opportunity for upward mobility is seriously constrained for lower-income families and that lack of opportunity has its own negative consequences for the educational aspirations of youth (Chetty et al., 2022; Kearney & Levine, 2014). Yet each of these literatures is largely nonoverlapping. Researchers tend to focus on opportunities within a particular setting or during a single stage of life, such as early childhood or middle childhood or adolescence.

The result is that our collective knowledge of opportunity gaps has primarily been pieced together from many empirical snapshots of a single aspect of opportunity during a single stage of children's development. Although theorists have worked to piece together these fragmented literatures, even basic descriptive questions remain difficult for the field to answer. For example, how large are cumulative opportunity gaps between the rich and poor? Many suspect that sizable opportunity gaps are accruing across settings and time (e.g., Carter & Welner, 2013), but direct evidence is needed. A convincing test of the opportunity gap hypothesis requires that children from a range of economic backgrounds be followed across settings and time while directly measuring their learning and growth opportunities and life outcomes.

The Present Study

In the present study, we investigated the accumulation of life opportunities, across early childhood, middle childhood, and adolescence, as a predictor of educational attainment and earnings at age 26 in a socioeconomically diverse sample living in the United States. It is the first investigation we are aware of to document cumulative opportunities accrued via caring, enriched environments in several key settings from early childhood through adolescence. To do so, we use data from a prospective longitudinal study in which children from lower-, middle-, and

higher-income families were followed from birth through age 26 with frequent, “gold standard” measurements of their developmental contexts and experiences from early childhood through adolescence. Using these data, we capture children’s exposure to opportunities in five key settings (home, early care and education, school, afterschool, and neighborhood) across three developmental periods (early childhood, middle childhood, adolescence) and opportunities of upward economic mobility during this time.

In our analyses, we addressed four research questions. First, we examined the number of opportunities experienced by children born into families with varying levels of incomes, hypothesizing that children born into low-income households would accumulate fewer opportunities—across settings and time—than children born into middle-income households, who, in turn, would accumulate fewer opportunities than those born into higher-income households. Second, we examined whether the accumulation of life opportunities between early childhood and adolescence was predictive of educational attainment and annual salary at age 26. We hypothesized that children who accumulate relatively more opportunities would evidence higher levels of educational attainment and earnings in early adulthood.

Third, we examined a mediation hypothesis. Specifically, we predicted that disparity in opportunity between children born into lower- versus higher-income households would mediate—help explain, statistically—associations between household income in early childhood and adult outcomes. Fourth, we examined whether associations between cumulative opportunities and outcomes in early adulthood varied by early childhood income. This last question was motivated by evidence that opportunity can be more strongly associated with achievement for children from low-income families than for their higher-income peers (e.g., Bustamante et al., 2022; Sorhagen, 2013; Wodtke et al., 2016). As such, we hypothesized that the accumulation of opportunity would be particularly key to the educational and financial successes of children born into low-income households.

Method

Data for the present study came from the National Institute of Child Health and Human Development (NICHD) Study of Early Child Care and Youth Development (SECCYD). Beginning in 1991, the SECCYD followed a birth cohort of 1,364 children living in or near 10 sites in the United States (Little Rock, AR; Irvine, CA; Lawrence, KS; Boston, MA; Philadelphia, PA; Pittsburgh, PA; Charlottesville, VA; Seattle, WA; Morganton, NC; and Madison, WI). Data on the children and their families, childcare arrangements, schools, afterschool experiences, and communities were collected repeatedly using a variety of different measurement tools (e.g., parent reports of family income; standardized assessments of parent personality and intelligence; researcher observations of homes, childcare settings, and school classrooms; and self-reports of educational and financial status in early adulthood). Of the original SECCYD sample, 814 remained in the study through age 26. These 814 were our analytic sample for all models presented here.

Measures

In the following, we provide brief summaries of our study variables. Additional details on measures are in the supplemental material (available on the journal website), including summary statistics and methods for handling missing data.

Income. Early childhood income in the present study was based on average family income-to-needs, a ratio calculated from family reports of annual household income (collected when children were 6, 15, 24, 36, and 54 months of age) divided by U.S. census poverty thresholds for the appropriate family size. An income-to-needs ratio of 1.0 denotes the poverty level, and 2.0 is considered the threshold for low-income (Bustamante et al., 2022; Dearing et al., 2009). In the present study, we defined low-income as income-to-needs ≤ 2.0 , middle-income as income-to-needs > 2.0 to ≤ 4.0 , and higher-income as income-to-needs > 4.0 . Of the participants who remained in the study through age 26, the average income-to-needs during early childhood was 3.69 ($SD = 2.89$), with 27.7% of the sample ($n = 226$) being low-income and nearly half of those children living below the poverty line ($n = 110$).

Cumulative opportunity. Our conceptualization of opportunities follows the assertion of systems theorists that the most powerful environmental influences on children come through proximal processes in settings that directly contain the child (Bronfenbrenner & Morris, 2007). We focus on home, early care and education, school, afterschool, and neighborhood, an approach justified by robust empirical literatures linking each of these contexts to achievement in the short and long terms (Bustamante et al., 2022; Dearing & Tang, 2010; Leventhal & Brooks-Gunn, 2000; Pianta et al., 2008; Vandell et al., 2007). In addition, we argue that families gaining or sustaining higher income status—across later childhood and adolescence—is an opportunity in and of itself.

To capture the accumulation of opportunities across contexts and developmental periods, we followed classic approaches to capturing cumulative risk (e.g., Felitti et al., 1998; Sameroff et al., 1998) whereby the total number of developmental risks to which children are exposed is summed. In our case, however, we summed exposure to exceptionally supportive contexts rather than exceptionally risky contexts. This approach is in line with momentum in the fields of education and psychology to take more asset-based framings when studying children and families from underresourced communities (e.g., Bustamante & Hindman, 2020) and was justified based on child development theory (Bronfenbrenner & Morris, 2007; Elder et al., 2003; Lerner & Castellino, 2002; Piaget & Cook, 1952; Scarr & McCartney, 1983).

Specifically, we created a 12-point cumulative opportunity index from a selection of well-established measures of the home, childcare, school, afterschool, and neighborhood environments that children experienced in early childhood (i.e., between 6 and 54 months of age), middle childhood (i.e., between kindergarten and fifth grade), and adolescence (i.e., between sixth grade and the end of high school). These measures included observations of home environments, observations of early childhood care and

education (ECE) quality, observations of elementary school classroom quality, maternal and youth reports of organized extracurricular and afterschool activities, U.S. census data on neighborhood conditions, and family-reported income-to-needs after early childhood. In the following, we first describe the overall process for creating our opportunity index and then describe the individual measures that were included in the index. More details on these measures and how we used them to construct our opportunity index are provided in the supplemental material (available on the journal website).

In total, our index was composed of the following 12 opportunities, ordered according to the developmental stages of early childhood, middle childhood, and adolescence. From measures collected during early childhood, we included three opportunities in the index: high-quality home environment, sustained time in high-quality ECE, and concentrated neighborhood advantage across the ages of 6 and 54 months. From measures collected during middle childhood, we included five opportunities in the index: high-quality home environment, high-quality elementary school classroom environment, sustained participation in organized afterschool activities, concentrated neighborhood advantage, and high family income-to-needs from kindergarten to fifth grade. From measures collected during adolescence, we included four opportunities in the index: high-quality home environment, sustained participation in organized afterschool activities, concentrated neighborhood advantage, and high family income-to-needs from sixth grade through the end of high school.

For each of these 12 opportunities, we used the average of multiple measurements that were conducted within developmental stages (e.g., in early childhood, quality of the home environment was based on the average of assessments conducted when children were 6, 15, 36, and 54 months of age). Next, from these average scores, we dichotomized children's experiences into those that were either above or below an opportunity threshold. The dichotomized indicators were then summed, yielding a 12-point index.

For two opportunity domains (i.e., ECE and family income-to-needs), thresholds for dichotomization were based on conceptually meaningful cut points that have been identified in the literature (see details in supplemental material, available on the journal website). In all other cases, we used the top 25% of the empirical distribution to identify opportunity thresholds. This approach followed (a) theory emphasizing the exceptional developmental impact of positive experiences and environments that are themselves exceptional in that they exceed the average expectable environment (Scarr, 1992), (b) a long line of work using the bottom or top 25% of empirical distributions for identifying risky experiences (e.g., Gach et al., 2018), and (c) strengths-based theoretical perspectives on the importance of identifying contexts that hold exceptional promise for promoting positive development (e.g., Lerner et al., 2005).

Turning to the measures of these opportunities, quality of the home environment was measured using the Home Observation for Measurement of the Environment (HOME; Bradley & Caldwell, 1984) four times during early childhood (i.e., 6, 15, 36, and 54 months of age), twice during middle childhood (i.e., third and fifth grades) and once in adolescence (i.e., 15 years of age). For

opportunity in ECE, we used scores from the Observational Record of the Caregiving Environment (ORCE; NICHD Early Child Care Research Network, 1996). Developed for the SECCYD, the ORCE is a live observational instrument that was used to assess the quality of ECE settings when children were 6, 15, 24, 36, and 54 months of age, including assessments of caregiver sensitivity and stimulation of cognitive development.

Classroom environment quality during elementary school was captured at first, third, and fifth grades using the Classroom Observation System (COS; Pianta et al., 2008). The COS includes multiple 7-point qualitative ratings that assess academic and socioemotional aspects of the classroom, including instructional quality, the academic enrichment of classroom activities, teacher sensitivity, and the emotional climate of the classroom. To capture children's participation in organized out-of-school activities (e.g., academic enrichment, clubs, lessons, sports, religious activities) during middle childhood, mothers were interviewed 15 times between kindergarten and fifth grade. In addition, at sixth grade, at age 15 years, and at the end of high school, adolescents self-reported involvement in organized afterschool activities.

Using 1990 and 2000 U.S. census data at the block level, our neighborhood opportunity indicators were created based on family reports of their home addresses. Specifically, we followed classic neighborhood research (e.g., Leventhal & Dupéré, 2019) on demographic indicators of concentrated neighborhood advantage by forming a composite of three values: (a) the percentage of families with annual incomes greater than \$100,000, (b) the percentage of household heads with a bachelor's degree or higher education, and (3) the percentage of individuals with professional or managerial jobs.

Our final two variables included in the opportunity index captured opportunities related to family income in middle childhood and adolescence. With early childhood income being our primary predictor of opportunity gaps and age 26 outcomes, we assume that (a) reaching or maintaining middle to high income in later developmental stages (e.g., a low-income family that experiences economic gains across a child's life) could offer children opportunities (e.g., quality health care) not captured by our other measures and (b) we should also expect that children born into low-income households would have less chance than other children to experience these opportunities given well-established constraints on their economic mobility. Specifically, we categorized children as experiencing an opportunity if their average family income-to-needs ratio was 3.0 or greater during these two times. With income-to-needs values of 2.0 or lower considered low-income and those between 2 and 3.0 considered "near low-income," an income-to-needs of 3.0 has been used in the literature as a threshold for middle income (e.g., Dearing et al., 2009).

Educational attainment and earnings. Educational attainment at age 26 was measured with the question, "What is the highest degree you earned?" We categorized response options ranging from "no high school diploma" to "doctoral degree (e.g., PhD, MD, EdD)" into four levels: high school or less, some college or a 2-year degree, 4-year degree, and greater than a 4-year degree. We also examined participants' self-reported annual salary at age 26.

Covariates. In our statistical models, we adjusted for child gender, ethnicity, and birth order and maternal age, education, partner status, attitudes toward employment, extraversion, agreeableness, neuroticism, and verbal intelligence.

Analytic Strategy

Preliminary analyses: the distribution of opportunities. As a preliminary step in our analytic plan, we descriptively examined the distribution of opportunities across family economic strata. These descriptive analyses included plotting (a) the total number of opportunities children experienced (i.e., histograms and fractional polynomial) and (b) the proportion of children who experienced each individual opportunity (i.e., heat plot) according to their early childhood family income.

Research Questions 1 to 4. Building on the descriptive analyses, we then addressed our four research questions by estimating a series of linear (ordinary least squares) and nonlinear (ordinal logit) regression models to examine direct, indirect (i.e., mediated), and moderated relations between early childhood income, cumulative opportunities, and age 26 outcomes. Specifically, to examine whether early childhood family income is associated with the number of opportunities children accumulate (Research Question 1), we estimated a model with early childhood income as a predictor of cumulative opportunities. Next, to examine whether cumulative opportunities are associated with age 26 educational attainment and earnings (Research Question 2) and whether cumulative opportunities help explain—mediate—links between early childhood income and age 26 outcomes (Research Question 3), we estimated early childhood income as a predictor of the age 26 outcomes prior to and after including cumulative opportunities as a predictor of these outcomes. Then, the mediation hypothesis (Research Question 3) was directly tested via the product of the coefficients comprising the indirect effect in these regressions (i.e., early childhood income $\rightarrow$ cumulative opportunities $\rightarrow$ age 26 outcomes), with confidence intervals estimated via bias-corrected bootstrapping (MacKinnon & Fairchild, 2009).

In each of the regression models for our first three research questions, we examined two specifications of early childhood income, a linear specification (i.e., opportunities and age 26 outcomes regressed on income-to-needs levels) and a semi-log specification (i.e., opportunities and age 26 outcomes regressed on log income-to-needs). The linear specification assumed a constant strength of association across levels of income-to-needs, and the semi-log specification allowed for stronger associations at the low end of the income distribution than at the high end.

Finally, we examined the moderating effects of early childhood income for the association between cumulative opportunities and age 26 outcomes (Research Question 4) by regressing educational attainment and earnings on the main effects of early childhood income (linear specification) and cumulative opportunities and the interaction of these two predictors.

Robustness and sensitivity. As a final step in our analytic plan, we conducted robustness checks for all regression models examining direct, indirect, and interaction effects. A primary goal of this

last step was to examine the robustness of our results when using alternative specifications of our cumulative opportunities variable and alternative specifications of the age 26 outcomes. In addition, we examined the sensitivity of all regression model results to omitted variable bias.

Results

Distribution of Opportunities

As displayed in Figure 1, stark differences in the total number of opportunities that children experienced as a function of early childhood income are evident. Whereas the modal number of opportunities was 1 for those with low-income early childhoods, the modal number of opportunities was 4 for those from middle-income early childhoods and 8 for those from higher-income early childhoods. Moreover, although fewer than 20% of low-income children had four or more opportunities, more than 66% of middle-income children and more than 90% of higher-income children had four or more opportunities.

We also present a heat plot (Figure 2) depicting the proportion of children who experienced each of the 12 opportunities assessed in our opportunity index; proportions are displayed as a function of income-to-needs decile in early childhood. As expected, children's likelihoods of experiencing opportunities appear, in general, greatest in the highest income-to-needs deciles. Interestingly, however, two opportunity domains demonstrated somewhat greater equity in likelihood of opportunity across family income levels: high-quality elementary school classrooms and organized afterschool activities (particularly in adolescence). In these two domains, children's likelihood of opportunity was near 25% or higher even among children in some of the lowest income households in early childhood.

Early Childhood Income, Cumulative Opportunities, and Age 26 Outcomes

In Table 1, we summarize the results of the regression models examining direct and indirect (mediated) relations between early childhood income, cumulative opportunities, and the age 26 outcomes. For our question of how strongly early childhood income was associated with cumulative opportunities, the linear income estimate indicated that on average, each 1-point gain in income-to-needs was associated with a .29-point increase in number of opportunities. However, the semi-log income estimate was considerably larger, indicating that each additional 1-point gain in income-to-needs was associated with a 1.25-point increase in opportunities for children in the lowest income households during early childhood. Taken together, the results from the linear and semi-log specifications are consistent with the fractional polynomial in Figure 1d: Disparities in cumulative opportunity were largest when comparing the lowest income families to those with marginally more income, with diminishing returns to income for those at the highest end of the income distribution.

A nonlinear relation was also evident for associations between early childhood income and age 26 outcomes. Prior to controlling for opportunities in our models—Columns 2 (for educational attainment) and 4 (for earnings) of Table 1—early

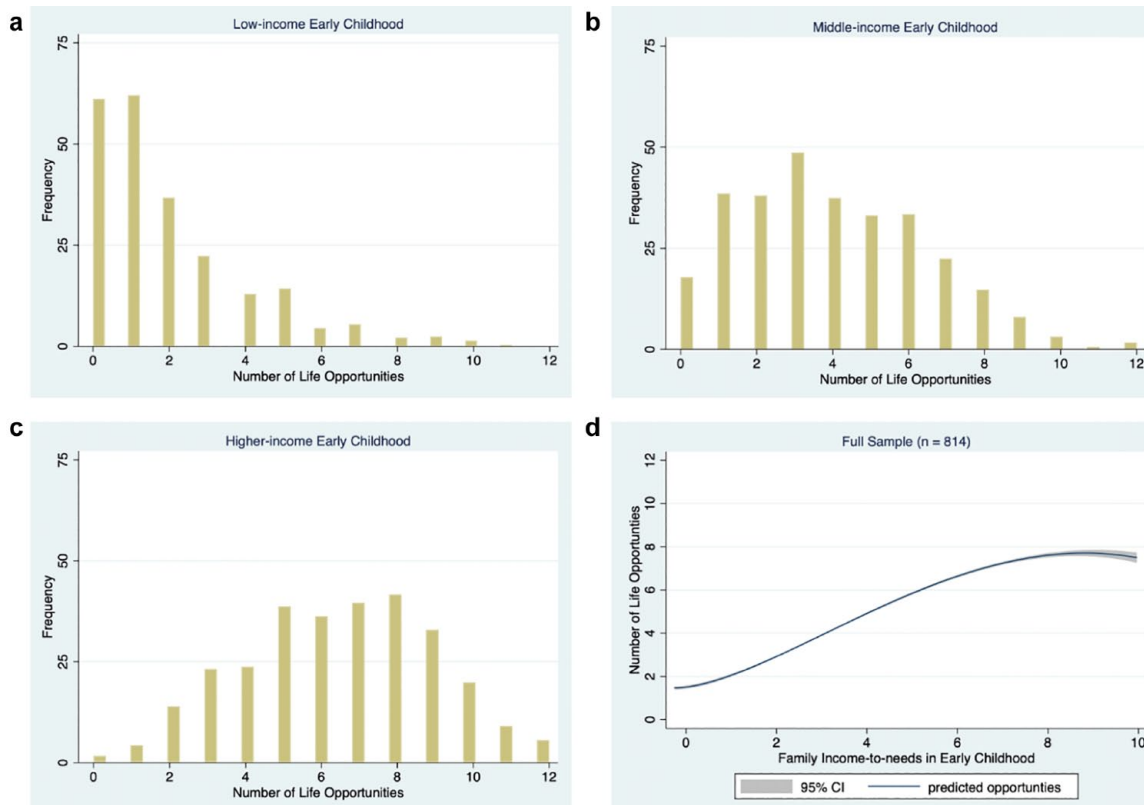


FIGURE 1. *Histogram plots of cumulative opportunities.*

Note. Distribution of cumulative opportunities for children who lived in (Fig. 2a) low-, (Fig. 2b) middle-, and (Fig. 2c) higher-income households during early childhood. In Figure 2D, the best-fitting fractional polynomial for the association between early income-to-needs and cumulative opportunities.

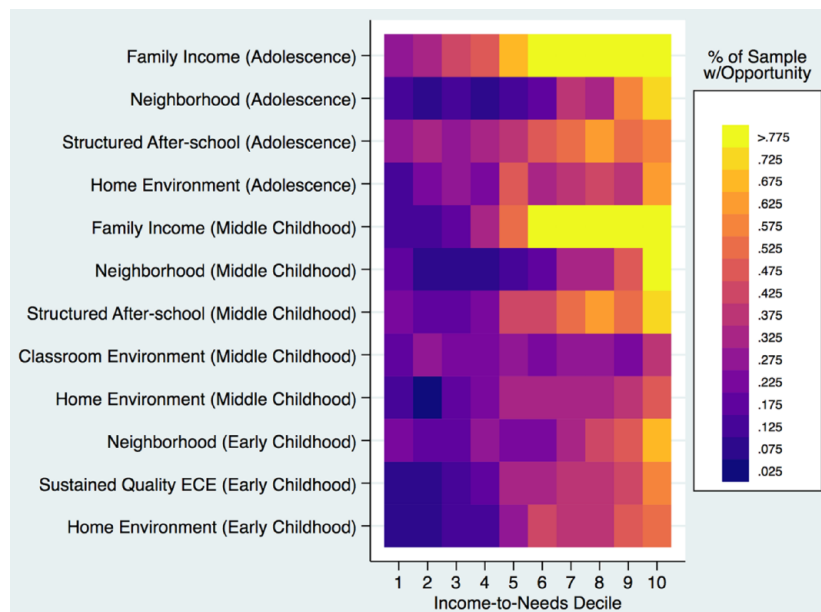


FIGURE 2. *Heat plot of opportunities by early childhood income.*

Note. Color variations indicate percentage of children, within income deciles, that experienced each opportunity.

childhood income was positively associated with level of educational attainment and earnings, and these associations were considerably larger in the semi-log models. For example, for children

from the poorest households, a 1-point gain in income-to-needs during early childhood predicted a \$6,800 increase in annual salary at age 26 (i.e., semi-log coefficient of $.68 \times 10,000$). Yet

Table 1
Early Income, Cumulative Opportunities, and Early Adult Outcomes: Estimated Direct and Mediated Effects

	Cumulative Opportunities	Educational Attainment		Age 26 Salary	
	<i>b</i> (SE)	<i>b</i> (SE)	<i>b</i> (SE)	<i>b</i> (SE)	<i>b</i> (SE)
Direct effects					
Early income-to-needs: linear estimate	.29*** (.04)	.06 (.03)	-.01 (.03)	.14** (.05)	.08 (.05)
Early income-to-needs: semi-log estimate	1.25*** (.19)	.38* (.16)	.11 (.17)	.68** (.23)	.42 (.22)
Cumulative opportunities			.24*** (.04)		.21*** (.05)
Mediated effects					
Linear income-to-needs → opportunities			.07***		.06***
Semi-log income-to-needs → opportunities			.30***		.26***

Note. Model covariates included child gender, race/ethnicity, and birth order; maternal age, education, verbal intelligence, extraversion, neuroticism, agreeableness, attitudes toward employment, and partner status; and study site. The linear and semi-log specifications of income-to-needs were estimated in separate models for each outcome. The mediation hypothesis was tested using the product of coefficients methods with bootstrapping for standard errors.

* $p < .05$. ** $p < .01$. *** $p < .001$.

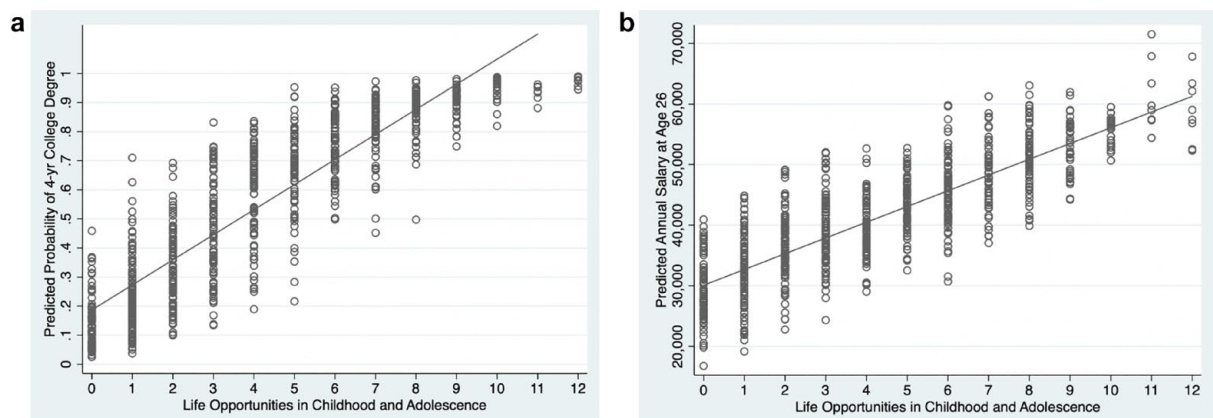


FIGURE 3. Predicted (a) probability of 4-year college degree and (b) annual salary as a function of cumulative opportunities, adjusted for early childhood income and child and family covariates.

once we included cumulative opportunities as predictors of the age 26 outcomes—Columns 3 (for educational attainment) and 5 (for earnings) of Table 1—two additional key findings were evident.

First, with both early childhood income and cumulative opportunities in the models, cumulative opportunities were strongly associated with educational attainment and earnings. In Figure 3a, for example, we display the predicted probability of gaining a 4-year college degree as a function of opportunities: The probability for those with no opportunities was less than 20%, and it was greater than 50% for those with at least four opportunities. For annual salary (Figure 3b), each additional opportunity was associated with a gain of approximately \$2,100. Second, links between early childhood income and the age 26 outcomes were largely attenuated in the models that included cumulative opportunities. Specifically, among the statistically significant associations between early childhood income and the

age 26 outcomes, coefficients were between 1.62 (i.e., semi-log income and earnings) and 3.46 (i.e., semi-log income and educational attainment) times smaller once cumulative opportunities were controlled in the models. Moreover, for both educational attainment and earnings, cumulative opportunities significantly mediated—helped account for—associations between early childhood income and the age 26 outcomes.

Moderating Effects of Early Childhood Income?

In addition to these mediation effects, associations between cumulative opportunities and both educational attainment and earnings were moderated by early childhood family income (Table 2). Opportunities were most strongly associated with the age 26 outcomes for those who lived in low-income households during early childhood; the association was less strong for children who grew up in higher-income households (Figures 4a and

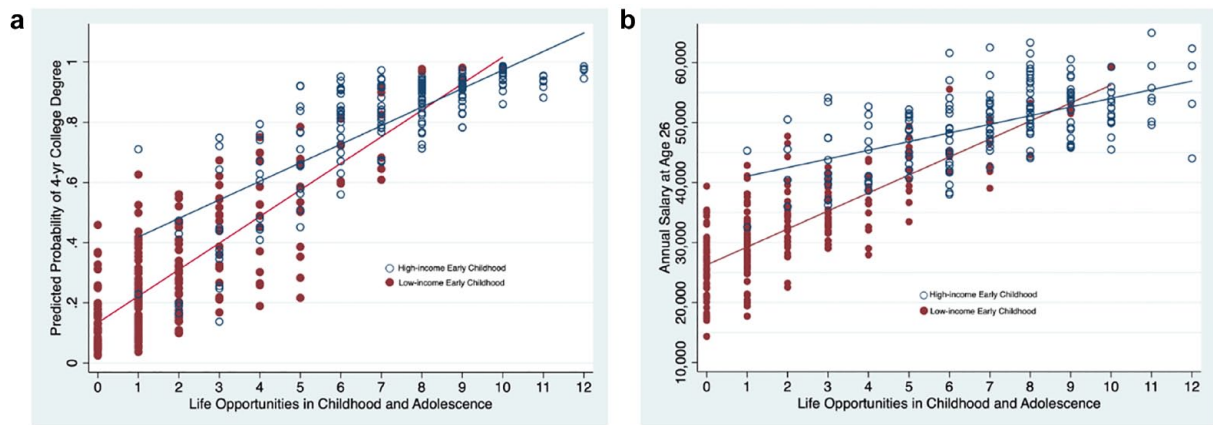


FIGURE 4. Predicted (a) probability of 4-year college degree and (b) annual salary as a function of cumulative opportunities and early childhood income, adjusted for child and family covariates.

Table 2
Opportunities Moderated by Early Childhood Income

	Educational Attainment	Age 26 Salary
	<i>b</i> (SE)	<i>b</i> (SE)
Early income-to-needs: linear estimate	.16 [†] (.08)	.23** (.09)
Opportunities	.33*** (.06)	.29*** (.06)
Income-to-Needs × Opportunities	-.02* (.01)	-.02* (.01)

Note. Model covariates included child gender, race/ethnicity, and birth order; maternal age, education, verbal intelligence, extraversion, neuroticism, agreeableness, attitudes toward employment, and partner status; and study site.
[†] $p < .10$. * $p < .05$. ** $p < .01$. *** $p < .001$.

4b). For example, for those born into low-income households, each additional opportunity was associated with about a 10% increase in the likelihood of graduating from a 4-year college; for those born into higher-income households, the association was about half that size. For earnings, each additional opportunity was also associated with gains nearly twice as large for those from low-income versus high-income early childhoods, approximately \$2,800 versus \$1,600 annually.

Robustness and Sensitivity

As a final step in our analyses, we conducted robustness checks that were aimed at understanding the impact of (a) alternative operationalizations of the opportunity index, (b) alternative operationalizations of the outcome variables, and (c) nonlinearity in the relations between opportunities and the outcomes. For example, for alternative operationalizations of the opportunity index, we formed (a) a 38-point index that treated each observation/assessment within home, childcare, school, afterschool, neighborhood, and income domains as distinct opportunities

(e.g., home environment quality at 6, 15, 36, and 54 months were each dichotomized at the 75th percentile and treated as four distinct opportunities); (b) a weighted composite score created using a structural equation measurement model estimated with the 38 opportunity indicators; and (c) a 10-point index that did not include middle childhood and adolescent family income as opportunities. These first two strategies were taken to ensure that we were justified in collapsing the 38 opportunity indicators into the 12-point cumulative opportunity index. The third strategy was taken because although including middle childhood and adolescent family income in our index helped capture opportunities that were not directly measured (e.g., health care), there is the risk of inflating the size of associations between (a) early childhood income and the opportunity index and/or (b) the opportunity index and adult outcomes.

Across these alternative variable specifications, the results were highly consistent with those presented in Tables 1 and 2. The most notable differences between the respecified models and those presented in Tables 1 and 2 were evident for the 10-point opportunity index that excluded children's family income in middle childhood and adolescence. Compared with our original index, if we did not count middle childhood and adolescent income as opportunities, associations between (a) early childhood income and cumulative opportunities and (b) cumulative opportunities and age 26 salary were attenuated. Nonetheless, these associations remained large and highly statistically significant ($p < .01$). Moreover, when using the 10-point index, the weaker association between cumulative opportunities and age 26 salary was primarily evident among children from higher-income backgrounds and much less so among those from low-income backgrounds; each additional opportunity was, on average, associated with about \$2,500 additional salary at age 26 for children from low-income households (i.e., a 3.9% reduction compared with our original opportunity index) but only about \$900 additional salary for children from higher-income households (i.e., a 43.8% reduction compared with our original opportunity index).

We also ran sensitivity analyses examining the robustness of our estimates to potential omitted variable bias. Across the models presented in Tables 1 and 2, the most sensitive result (i.e.,

least robust) was the link between opportunities and annual salary, although even for this finding, considerable bias would be necessary to change our inference. For example, to nullify the association between cumulative opportunity and earnings at age 26, omitted variables need to explain 4 times more variance in earnings than did educational attainment at this age. Full details on all robustness and sensitivity analyses are provided in the supplemental material (available on the journal website).

Discussion

This study offers novel evidence that the accumulation of opportunities across contexts and developmental periods is key to understanding the long-term consequences of low family income during early life. In this U.S. sample, children born into low-income households often experienced very few, or none, of the educational opportunities that we assessed—at home, childcare, school, afterschool, or in the neighborhood—at any point between early childhood and adolescence. In turn, accumulating disparity in access to life opportunities between children from lower- versus higher-income backgrounds largely explained differences in their educational attainment and earnings at age 26. We also found that each additional opportunity was most strongly and positively associated with education and earnings in adulthood for those children who had accumulated the fewest chances to thrive.

Social science research has a history of highlighting the ways in which children from low-income families achieve at levels below their middle- and higher-income peers, often hypothesizing deficits, failures, and maladaptive practices within families as drivers of disparity. Our results suggest something broader and more systemic is occurring: Young children whose families have low income have dramatically fewer growth-promoting opportunities from early childhood through high school. These opportunity gaps largely account for differences in educational attainment and earnings of low-income children and their more affluent peers. Findings from the present study bolster opportunity gap hypotheses (Carter & Welner, 2013; Chetty & Hendren, 2018; Gorski, 2017; NASEM, 2023b) with critical implications for intervention approaches, albeit with important caveats for the correlational design of our study.

Relying on correlational data, we are mindful of selection bias concerns (e.g., children were not randomly assigned to opportunities). Yet the robustness of the associations in our study underscores their scientific significance. Moreover, our descriptive data offer insights into where intervention might build from family, school, and community assets in contexts of disadvantage. For example, there were deeper disparities in areas such as access to high-quality early education. Conversely, opportunity gaps were less stark in the domains of quality elementary education and afterschool programming. Some children from lower-income households in the SECCYD may have benefited from teacher quality improvement efforts across the 1990s (National Center for Education Statistics, 2003) or from increased afterschool programming in lower-income communities during the 2000s (e.g., California Department of Education, 2023; U.S. Department of Education, 2023). Although more mapping of opportunities across contexts and development could help target

investments toward domains with the largest gaps, two points are worth considering.

First, given sizable differences in opportunity that we find accumulating across settings and age, we are pessimistic that the net impact of piecemeal investments can reach that of more holistic investments in equity. With piecemeal policy approaches, the benefits of investments in one opportunity domain can be offset by increasing inequity in other domains. At the population level, achievement disparities between the advantaged and disadvantaged can remain stable or increase when the benefits of investments in opportunity domains such as education are washed out by rising inequality in other domains such as family income, wealth, and residential segregation (Dearing, 2023; Sandsør et al., 2023). Given we find that children from low-income households are growing up with many fewer opportunities that span five settings (home, early childhood programs, K–12 schools, afterschool programs, and neighborhoods) and three developmental periods (early childhood, middle childhood, adolescence), systemic investment in opportunities across contexts and developmental periods may be required to create lasting impacts.

Interventions that offer opportunities across time and settings may be part of the solution. Consider, for example, increasing policy support for school-based interventions alternatively referred to as “community school partnership” programs or “integrated student support” (e.g., Center for Thriving Children, 2023). With evidence of decreased high school dropout and increased college completion (Lee-St. John et al., 2018; Pollack et al., 2023), these interventions use schools as hubs for connecting low-income children and their families with a range of opportunities through partnerships with community-based agencies, from afterschool programming to housing assistance and health care and many other forms of enrichment and support.

Second, although the present study provides a major advance for opportunity gap hypotheses, the field needs more long-term longitudinal studies with direct assessments of children’s educational opportunities at home, childcare, school, afterschool, and in the community. Ideally, these studies would include both frequent and balanced timing of assessments across developmental stages; a limitation of the SECCYD was the varying number of measurements of opportunity domains during early childhood, middle childhood, and adolescence (e.g., no observations of high school classrooms). Recent—nationally representative—longitudinal data are available or being collected (e.g., Institute of Education Sciences, 2024), but few studies follow children across multiple developmental stages with the same quality of measurement as provided in the SECCYD (e.g., classroom observations). The SECCYD, however, began data collection in the early 1990s, when the racial and ethnic makeup of the country was substantially different. White families are overrepresented and Latine families underrepresented compared to the current demographic makeup of the United States. Thus, we assume some family and community strengths were missed in the present study, particularly among racially, ethnically, and linguistically minoritized groups.

Important changes in the opportunity landscape have also occurred over the last few decades in the United States, for better and for worse. For example, although investments in high-quality public preschool have risen in nearly every state over the last 3

decades, so too have rates of residential income segregation and school racial segregation (Reardon et al., 2018). In addition, the income distribution in the United States has changed in salient ways, including rising wages and decreasing poverty rates but also increasing income inequality and wealth inequality alongside rising consumer debt for middle-income families (Saez & Zucman, 2020; Sullivan et al., 2020). These changes could be distancing even further children's access to opportunities not only between the wealthy and poor but also between those living in wealthy and middle-income families (Chetty et al., 2024). With these contemporary changes in mind, the present study provides a major advance in accounting for children's opportunities across development and contexts using longitudinal data in novel ways to bring much needed cohesion and clarity to a fragmented evidence base.

Conclusion

This 26-year longitudinal study provides novel and robust evidence of the differential accumulation of opportunities in the United States for children in lower- and higher-income households. Although correlational, the study also points to the potential for serious consequences of differences in cumulative opportunities for educational attainment and earnings in adulthood. These findings are consistent with the opportunity gap hypothesis that systemic inequity in access to opportunities drives disparities in life outcomes for children from lower- and higher-income households.

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